Building a Comprehensive Music Dataset for Sentiment Analysis

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Data Management Project

Abstract
This project focuses on constructing a robust and comprehensive music dataset by integrating data from Spotify's API, Kaggle, and AZLyrics, laying the foundation for diverse analytical applications. The dataset preparation involved acquiring top global songs from Spotify (2020–2023), supplementing them with audio features from Kaggle, and enriching them with lyrics through web scraping from AZLyrics. Employing a relational database model ensures consistency, completeness, and scalability, making the dataset ready for extensive exploration. Potential analyses include sentiment, mood analysis and clustering songs by textual similarity. This foundational work provides a versatile resource for future studies into listener preferences, musical trends, and the dynamic interplay of lyrics and sound.

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1 Introduction

1.1 Data Understanding and Objectives

The music industry provides a captivating domain for exploration, reflecting cultural, social, and artistic trends that resonate with listeners worldwide. This project focuses on constructing a robust and comprehensive dataset that captures the essence of top global songs, enabling future analyses of musical and lyrical attributes. By preparing a structured database, we lay the groundwork for research into what makes songs popular, how they engage audiences, and the patterns behind their success.

The dataset is designed to integrate multiple dimensions of music. Audio features such as energy, danceability, and acousticness offer insights into listeners' auditory preferences. Song lyrics, on the other hand, provide a lens

to understand the emotional and thematic content of music, revealing the sentiments and moods that connect with audiences. Additionally, analyzing release dates opens opportunities to explore the seasonality of music, examining when songs are released and how timing impacts their success.

Rather than performing the analyses ourselves, this project focuses on preparing a high-quality dataset that adheres to the core principles of the data management lifecycle. The process involves defining objectives, collecting diverse data, cleaning and standardizing records, integrating datasets, and storing the results in a structured relational database for future use.

To overcome limitations in data availability from a single source, we used a multi-pronged approach:

- Spotify API: To retrieve metadata about the top 50 global songs for 2020, 2021, 2022, and 2023, including song titles, artists, release dates, and popularity metrics.
- Kaggle datasets: To supplement the Spotify data with additional audio features, such as danceability, valence, and energy, which are critical for understanding musical attributes.
- Web scraping from AZLyrics: To collect song lyrics, a crucial resource for sentiment and thematic analysis, unavailable in the previous sources.

By combining these sources, this project delivers a cohesive and enriched dataset that can support a variety of future analytical tasks, including sentiment analysis, listener preference studies, and trend exploration in the music industry.

1.2 Methodology

The project has been developed in Python. The steps required different libraries: ***Pandas*** was used to easily manage data, ***Spotipy*** was necessary to access and communicate with Spotify's API, ***BeautifulSoup*** and ***time*** have been fundamental for ***WebScraping***, ***Re*** has been used to clean many columns, while ***mysql.connector*** allowed us to connect to our SQL database and store the tables.

2 Data Acquisition

2.1 Spotify API

The data acquisition process began with Spotify's API, which served as the primary source for collecting information about the most popular global songs.
Authentication was required to access the API, which provided us with programmatic access to Spotify's database. Once authenticated, we identified and used specific playlist IDs for the "Top 50 Global Songs" playlists for the years 2020, 2021, 2022, and 2023.

The API allowed us to extract separate datasets for each year, each containing 50 rows representing the top songs for that year.
Each dataset shared the same structure and format, as they originated from the same source.
Specifically, the dataset consisted of the following columns:

- **ID:** Unique identifier for each track.
- **Name:** Title of the song.
- **Artist:** Name of the performing artist(s).
- **Album:** Album to which the song belongs.

- **Release Date:** Date when the song was released.
- **Genre:** Primary genre(s) associated with the song.

This consistent structure facilitated a straightforward integration of these datasets into our analysis pipeline.

However, while Spotify's API provided a wealth of information regarding song metadata, we encountered a significant limitation: Spotify no longer offers detailed insights into audio features (e.g., danceability, energy, valence) through their API.
This restricted our ability to analyze these crucial attributes directly. Consequently, we sought alternative sources, such as Kaggle, to supplement the missing data and ensure a comprehensive analysis.

2.2 Kaggle Dataset

On Kaggle, we directly downloaded datasets for the top 50 hit songs of each year. This decision was made because Kaggle provided datasets with interesting and important features that were essential for our analysis. Unlike Spotify, Kaggle datasets are not guaranteed to have a consistent structure or format across different years. This variability arises from the fact that datasets on Kaggle are often uploaded by various providers, each with their own structure and conventions. Therefore, even though we downloaded the top 50 hit songs for each year from the same platform, the datasets were not inherently homogeneous in structure or data types.

To address this, we carefully examined and verified that the columns relevant to our analysis were present in all the datasets. For example, while we initially considered downloading a much larger dataset from Kaggle to include more records, we ultimately decided against it. The reason was that such a dataset would not align with our focus on specific years and the objectives of our analysis. Ensuring coherence and consistency was prioritized over simply increasing the number of records.

The columns we identified as interesting and relevant to our analysis are as follows:

- **ID:** Unique identifier for each track.
- **artist:** Name of the performing artist(s).
- **album:** Album to which the song belongs.
- **track_name:** Title of the song.
- **track_id:** Identifier for the specific track within the dataset.
- **energy:** Measures the intensity and activity of the song; higher values indicate more energetic songs.
- **danceability:** Indicates how suitable the track is for dancing; higher values mean the track is more danceable.
- **key:** The key in which the track is composed, represented as integers (e.g., 0 = C, 1 = C/Db, etc.).
- **loudness:** Overall loudness of the track in decibels (dB); higher values indicate louder songs.
- **acousticness:** A confidence measure from 0.0 to 1.0 of whether the track is acoustic.
- **speechiness:** Detects the presence of spoken words; higher values indicate more spoken content in the track.
- **instrumentalness:** Predicts whether a track contains no vocals; higher values mean the track is more instrumental.
- **liveness:** Detects the presence of an audience in the recording; higher values indicate a live performance.
- **valence:** Describes the musical positiveness of a track; higher values mean

the track sounds happier and more positive.

- **tempo:** The speed of the track, measured in beats per minute (BPM).
- **duration_ms:** Length of the song in milliseconds.
- **genre:** Primary genre(s) associated with the song.

This approach ensured that our datasets remained consistent and aligned with the goals of our analysis, providing the foundation for further exploration and integration with other data sources.

2.3 Web Scraping Lyrics from AZLyrics

To retrieve the lyrics for the songs in our dataset, we used web scraping techniques on the **AZLyrics** website. This method was essential to supplement our analysis with the textual content of the songs, as neither Spotify nor Kaggle provided the lyrics to the songs. The process involved generating URLs dynamically for each song by identifying patterns in the AZLyrics URL structure. Specifically, we manually analyzed how the website constructs its URLs and found that it eliminates spaces and special characters like "?" or "!" while also following a specific format for artist names and song titles.

The code was designed to generate various URL patterns for each song, such as combining the artist's full name without spaces or using only parts of the artist's name, along with the title of the song formatted appropriately. For instance, a song by an artist named "Taylor Swift" with the title "Anti-Hero" would result in URL attempts like:

- `taylorswift/antihero`
- `taylor/antihero`
- `swift/antihero`

It has been fundamental to use a sleep function; in our case, an interval between 10 and 17 seconds was enough to avoid detection from the site.
Despite the structured approach, we encountered some challenges, where scraping did not yield results. A significant issue was that certain artists have alternate names on AZLyrics. For instance, some artist names required adjustments to match the URL patterns used on the AZLyrics website:

- `BTS` was adjusted to `bangtanboys`
- `Beyoncé` was adjusted to `beyonceknowles`

To address this issue, we implemented a dictionary-based adjustment in our code to map these artist names to their AZLyrics counterparts.
These adjustments were necessary to ensuring that the scraping returned accurate lyrics for songs associated with these artists.
The results of the web scraping were then incorporated into our dataset, significantly enriching the data with lyrical content.
This enrichment allowed further analysis of the emotional and thematic aspects of the songs, which are crucial for understanding listener preferences and trends in the music industry.

3 Data Storage

3.1 MySQL and the relational model

After acquiring all the data, we opted for the relational model as the foundation for our

database. This decision was driven by several reasons:

- Fixed table structures: The data we are working with has a well-defined schema, making it suitable for a relational model. The consistency of the data ensures that future updates and extensions will be seamless and efficient.
- Predictable updates: New records, such as top songs from subsequent years, are released annually. This predictable influx of data aligns well with the stability and update-friendly nature of relational databases.
- Lightweight joins: The relationships between tables are relatively straightforward, with a low number of joins required for typical queries. This simplicity ensures that the database performs efficiently, even as the dataset grows.
- Structured data types: Apart from the semi-structured lyrics data, all other variables (such as track attributes and metadata) are composed of well-structured data types, including strings, integers, and floats. This compatibility further reinforces the suitability of the relational model for our dataset.

We believe the best architecture for our database can be effectively captured in the E-R model presented in Figure 1 of the Appendix.

The integration of the four Spotify datasets was relatively straightforward due to their consistent format and structure, all originating from the same source, allowing us to seamlessly concatenate all the datasets to create a unified dataset spanning the years 2020, 2021, 2022, and 2023.

On the other hand, the integration of the four Kaggle datasets required significantly more adjustments. Even though all datasets were sourced from Kaggle, their varying structures and formats presented challenges. These differences arose due to the diverse ways the data was compiled and uploaded by different data provider. One of the key issues encountered was **Name Heterogeneity** (synonymy), where columns with identical meanings had different names across datasets. For example, columns representing the song title, artist, or audio features like danceability and energy were labeled differently in each dataset.

To address these inconsistencies, we began by identifying and standardizing the columns common across all datasets. Fortunately, the essential columns for our analysis—such as title, artist, album, genre, and audio features like danceability, energy, and loudness—were present in every dataset. These columns were renamed to maintain consistency, ensuring alignment across datasets.

A particular issue challenge arose with the Kaggle dataset from 2023, which lacked the crucial "ID" column—a unique identifier necessary for integration. To resolve this, we leveraged Spotify's API to generate the missing IDs by matching track titles and artist names in the 2023 dataset with Spotify's data. This ensured that all datasets shared a unique identifier, which was essential for preserving data integrity during integration.

However, we faced challenges in maintaining completeness and consistency during data integration. Conflicts between column names and formats across different datasets had the potential to lead to incomplete or inaccurate integrations, jeopardizing the reliability of subsequent analyses. These challenges underscored the importance of careful standardization and validation at each stage of the data preparation process.

By resolving these issues and leveraging the strengths of the relational model, we were able to establish a coherent and integrated dataset that serves as a reliable foundation for future analyses of musical trends and preferences.

4 Data Preparation

4.1 Data Integration

The goal of the integration was to obtain a table with the information from the Spotify table and the metrics contained in the four Kaggle datasets. Because we encountered a Fragmentation conflict we decided to proceed as follows: concatenate the four Kaggle dataset, suddenly merge the Spotify dataset with the new and complete Kaggle dataset.

The Kaggle datasets were combined by appending them together, resulting in a unified Kaggle dataset with consistent structure and format, ready for further analysis and comparison with the Spotify dataset.

The next step has been the merge of the two datasets, firstly by ID and suddenly by Track Name, obtaining about 100 records. Because of the lower number of records with respect to the initial datasets with 200 records, the idea was to update the database as the E-R model in Fugure 2 of the Appendix shows.

The table SPOTIFY NOT IN COMMON, and the related FEAT and LYRICS tables, contains only the information from Spotify'API of the songs the two data sources *don't* have in common. SPOTIFY AND KAGGLE is the 'official' dataset with all the informations and metrics of the songs officially in Top 50 of the four years. This structure allows us to derive the dataset with some of the official Top 50 songs from Spotify and their metrics, but at the same time to store the texts and the collaborations of the songs without the metrics, which can be useful and valid to do sentimental analysis or train models.

4.2 Data Cleaning

At this point we checked the tables that had to be cleaned. The main adjustment was to delete the information regarding the featuring from the songs title after storing them into the Featuring tables. The other problem we solved was to detect the duplicates and deleting them, but following the Minimization principle. We want to underline that none of the lyric we stored have been cleaned, for example by lemmatization or tokenization techniques, because we want to present them as raw as possible, allowing the user to decide the best way to procede. The same thinking has been done with the column ***Genre***, indeed all the songs have many genres and sub-genre associated: we could have scraped, for example, https://en.wikipedia.org/wiki/List_of_music_genres_and_styles and used the lists obtained to cluster the genres, but we opted to provide the highest granularity possible.

5 Data Quality assessment

5.1 Accuracy

Accuracy ensures that the data reflects the real-world information it is meant to represent. For this project, inconsistencies were identified in various datasets due to differences in naming conventions and missing identifiers. For example, discrepancies in song titles and artist names between Spotify and Kaggle were resolved using string matching techniques and manual verification when needed. Additionally, specific adjustments, such as standardizing artist names for web

scraping (e.g., mapping "BTS" to "bangtanboys"), were implemented to improve data accuracy. These measures ensured that the dataset closely mirrors the actual attributes of the tracks being analyzed.

5.2 Completeness

Completeness ensures that all necessary data fields are adequately populated to meet the requirements of our analysis. It measures the percentage of missing values in the dataset, evaluating how complete the data is.

Formula:

$$\%MissValues = \left(\frac{TotalMissValues}{TotalCells}\right) \times 100$$

Results:

- Dataset 1: 20.69% missing values.
- Dataset 2: 20.45% missing values.
- Dataset 3: 0.18% missing values.
- Dataset 4: 0.13% missing values.
- Dataset 5: 0.42% missing values.

The percentage of missing values is quite high in the first two datasets, which is irrelevant since it's normal that featuring_2 column is empty for some songs. The dataset 3 and 5 have few missing values, this means that the scraping algorithm can't find these songs on the web page: we decided to delete the songs deprived of the lyrics.

5.3 Pertinence

Pertinence evaluates whether the collected data aligns with the project's goals. In this study, only the top 50 global tracks from 2020 to 2023 were included, ensuring a focused scope relevant to analyzing music trends and listener preferences. Features like danceability, energy, and lyrics were chosen for their direct relevance to sentiment analysis and musical exploration, while extraneous data were excluded to streamline the dataset for its intended use.

5.4 Consistency

Consistency refers to the degree to which data adheres to a uniform format, structure, and logic across records, datasets, or systems. It ensures that the data remains uniform across different sources and datasets. For this reason, we significantly focused on harmonizing the column names, data types, and formats when merging Spotify and Kaggle datasets. This step was critical for avoiding errors during analysis and maintaining a cohesive dataset. Duplicates represent one important tool to evaluate the consistency of the dataset, as it indicates redundancy and inconsistency in it. In our project, we identified some duplicates in:

- Dataset 3: 7 duplicates.
- Dataset 5: 1 duplicate.

To address this problem, a deduplication method is needed, by eliminating to duplicated records in order to maintain only one. Often, the duplication happens because the same song can refer to different ID, since the artist could have released the same song multiple times, in this case the "release_date" feature can be helpful to choose which version the user prefers to use. After this approach, the result is as follows:

- Dataset 3: 0 duplicates.
- Dataset 5: 1 duplicate.

6 Data Exploration

The potential uses and analyses an user has at their disposal are multiple and different from each other. We present some analyses to which the data lend themselves:

- Basic EDAs, for example creating Violin plot of the genres, correlation analysis of the Variable Duration and other variables, time series of the number of songs with the same genre or artist;
- Check if the most popular artists are 'multi-genre' or they obtain success focusing on only one genre;
- Basic analyses on the lyrics, for example: most used words based on genre, mean lenght of the texts;
- Sentimental analysis and creation of the concept of 'mood' based on the lexicon used (sad-related words or happy-related words);
- Correlation analysis between 'mood' and sounds metrics like Danceability, Loudness and Energy;
- Clustering of the songs based on similar lyrics;
- Explore NLP models to create lyrics based on which characteristics we desire.

7 Conclusion

This project demonstrates the effectiveness of integrating diverse data sources to create a unified and enriched dataset for the music industry. By combining structured metadata from Spotify, additional audio features from Kaggle, and lyrical content via web scraping from AZLyrics, we provide a resource capable of supporting in-depth analyses of song characteristics and listener engagement. Employing a relational database model ensured the data's consistency and scalability, addressing challenges like name heterogeneity and missing identifiers during integration. While this project focuses on data preparation and storage rather than analysis, the dataset is designed to facilitate future studies, such as sentiment analysis, trend exploration, and audience behavior modeling. This work underscores the importance of rigorous data management practices in constructing datasets that can drive meaningful insights and applications in the music industry.

8 Appendix

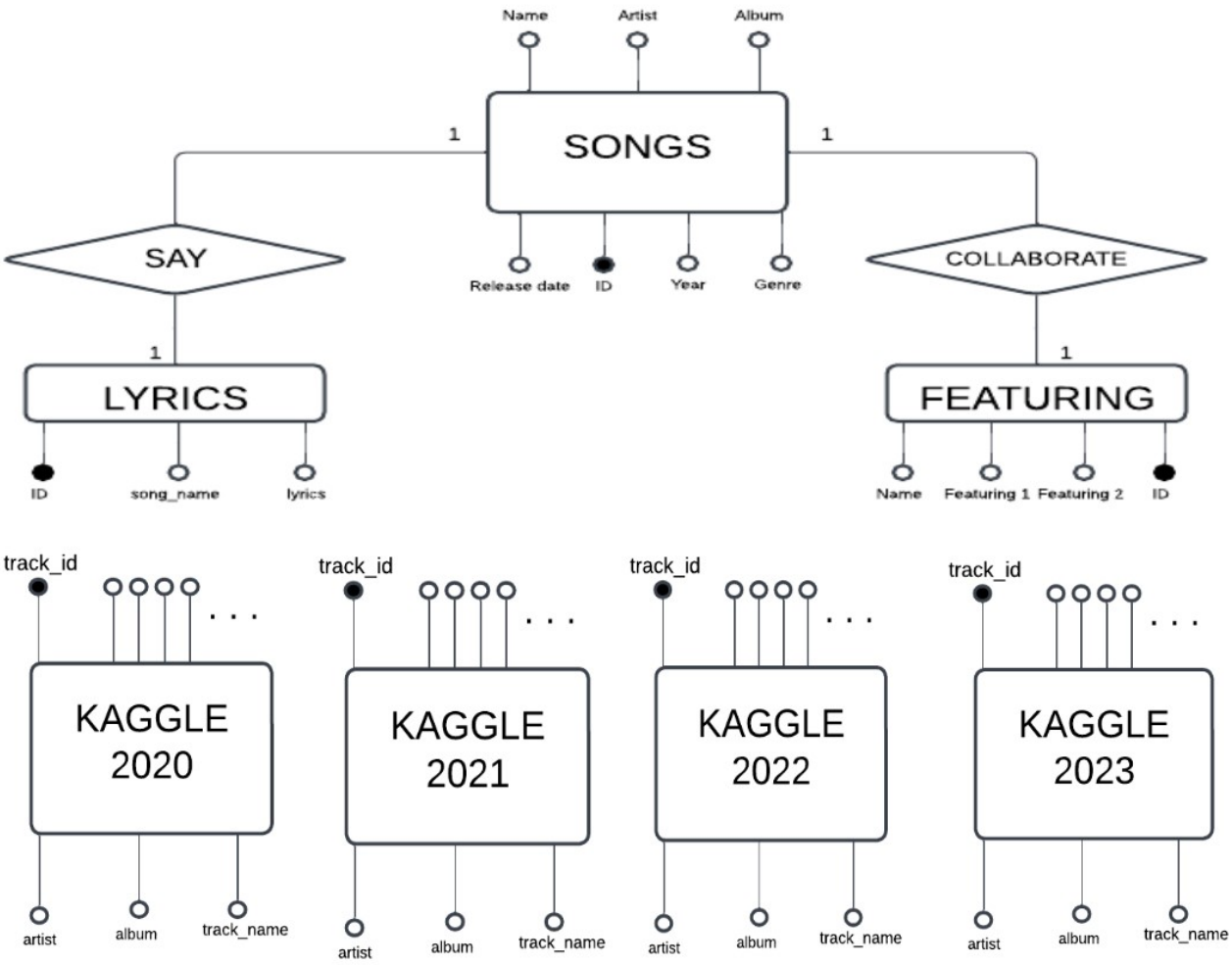

Figure 1: E-R model of the tables after the Acquisition

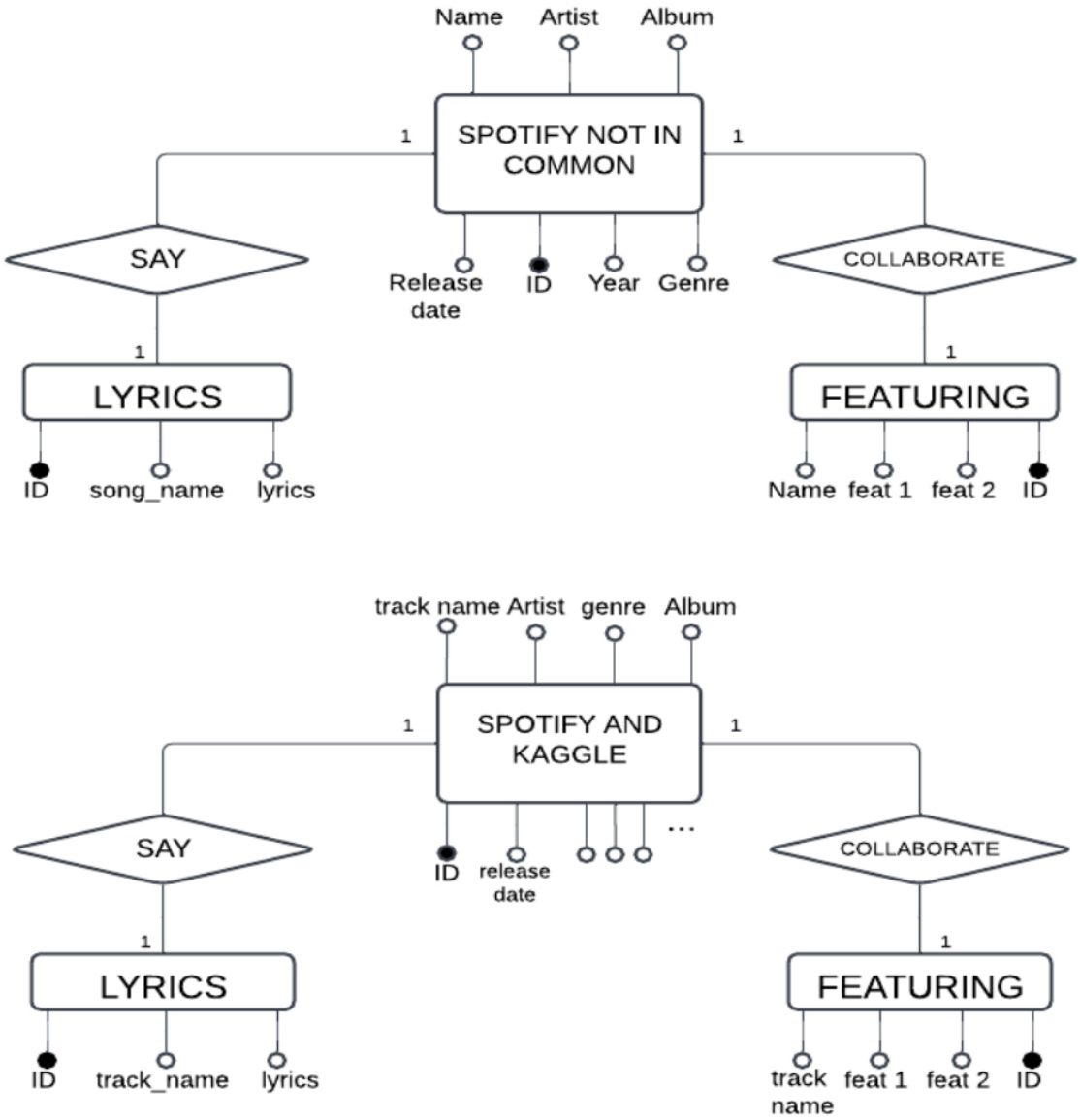

Figure 2: E-R model of the tables after the Integration